WORKED SOLUTION

STEP II 1992, Question 9

Pure • Scalar Triple Product

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q9

The key idea

Rearranging the expression for $\mathbf{r}$ as $\mathbf{a}$ plus combinations of $\mathbf{b}-\mathbf{a}$ and $\mathbf{c}-\mathbf{a}$ shows it is a plane through A, B and C. A normal to that plane is the cross product of two vectors lying in it, and expanding that cross product produces exactly the three triple products in the question. After that, the identity holds for *every* λ and μ, so we may choose convenient values.

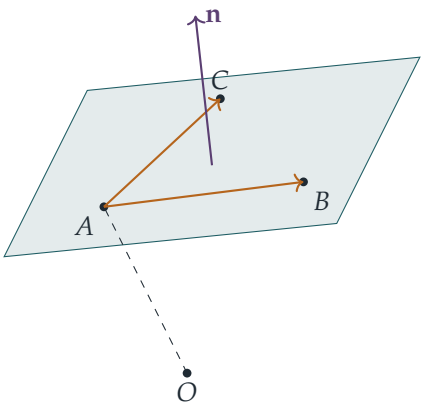
Solution

The locus

Rewrite the equation as

$$\mathbf{r} = \mathbf{a} + \lambda(\mathbf{b}-\mathbf{a}) + \mu(\mathbf{c}-\mathbf{a}).$$

This is the point A plus any combination of the vectors $\overrightarrow{AB} = \mathbf{b}-\mathbf{a}$ and $\overrightarrow{AC} = \mathbf{c}-\mathbf{a}$. These two vectors aren't parallel: if they were, A, B, C would lie on a line, and a line together with O always lies in some plane, which is ruled out. So as λ and μ range over all real numbers, $\mathbf{r}$ sweeps out the **plane through** A**,** B **and** C. Taking $(\lambda,\mu) = (0,0)$, $(1,0)$, $(0,1)$ gives A, B, C themselves. Since O is not in the plane ABC, this plane does not pass through the origin.



The equation $\mathbf{r}\cdot\mathbf{n}=p$

A normal to the plane is perpendicular to both $\overrightarrow{AB}$ and $\overrightarrow{AC}$, so we take

$$\mathbf{n}=(\mathbf{b}-\mathbf{a})\times(\mathbf{c}-\mathbf{a}),$$

which is nonzero because the two vectors are not parallel. Expanding, and using $\mathbf{a}\times\mathbf{a}=\mathbf{0}$ and $\mathbf{u}\times\mathbf{v}=-\mathbf{v}\times\mathbf{u}$,

$$\begin{aligned}\mathbf{n}&=\mathbf{b}\times\mathbf{c}-\mathbf{b}\times\mathbf{a}-\mathbf{a}\times\mathbf{c}+\mathbf{a}\times\mathbf{a}\\&=\mathbf{b}\times\mathbf{c}+\mathbf{c}\times\mathbf{a}+\mathbf{a}\times\mathbf{b}.\end{aligned}$$

Every point of the plane differs from A by a vector perpendicular to $\mathbf{n}$, so $\mathbf{r}\cdot\mathbf{n}=\mathbf{a}\cdot\mathbf{n}$. Since $\mathbf{a}$ is perpendicular to $\mathbf{c}\times\mathbf{a}$ and to $\mathbf{a}\times\mathbf{b}$,

$$p=\mathbf{a}\cdot\mathbf{n}=\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}).$$

So the plane is $\mathbf{r}\cdot(\mathbf{b}\times\mathbf{c}+\mathbf{c}\times\mathbf{a}+\mathbf{a}\times\mathbf{b})=\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})$.

? **Ask yourself** What is $\mathbf{r}\cdot\mathbf{n}$ when we substitute the parametric form of $\mathbf{r}$?

We need $\mathbf{a}\cdot\mathbf{n}$, $\mathbf{b}\cdot\mathbf{n}$ and $\mathbf{c}\cdot\mathbf{n}$. In each, two of the three terms vanish, because a vector is perpendicular to any cross product that contains it:

$$\mathbf{a}\cdot\mathbf{n}=\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}),\qquad\mathbf{b}\cdot\mathbf{n}=\mathbf{b}\cdot(\mathbf{c}\times\mathbf{a}),\qquad\mathbf{c}\cdot\mathbf{n}=\mathbf{c}\cdot(\mathbf{a}\times\mathbf{b}).$$

The point $\mathbf{r}=(1-\lambda-\mu)\mathbf{a}+\lambda\mathbf{b}+\mu\mathbf{c}$ lies in the plane for every λ, μ, so $\mathbf{r}\cdot\mathbf{n}=p$ gives

$$(1-\lambda-\mu)\,\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})+\lambda\,\mathbf{b}\cdot(\mathbf{c}\times\mathbf{a})+\mu\,\mathbf{c}\cdot(\mathbf{a}\times\mathbf{b})=\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}).$$

Subtracting $\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})$ from both sides,

$$-(\lambda+\mu)\,\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})+\lambda\,\mathbf{b}\cdot(\mathbf{c}\times\mathbf{a})+\mu\,\mathbf{c}\cdot(\mathbf{a}\times\mathbf{b})=0$$

for all scalars λ, μ, as required.

The deduction

Since the identity holds for all λ and μ, choose values that isolate terms.

- $\lambda = 1$, $\mu = 0$: $-\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) + \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) = 0$.
- $\lambda = 0$, $\mu = 1$: $-\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) + \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}) = 0$.

Together these give

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) = \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}).$$

(Equivalently, the left-hand side is linear in λ and μ and identically zero, so the coefficients of λ and of μ must both vanish.)

When A, B, C and O are coplanar

Then $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ all lie in one plane through O. Now $\mathbf{b} \times \mathbf{c}$ is perpendicular to that plane (or zero), so $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 0$, and likewise the other two triple products vanish. Geometrically, the parallelepiped on $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ is flat and has zero volume. The result is still true, but trivially: $0 = 0 = 0$. If A, B, C are not collinear the locus is still a plane, but it passes through the origin, so $p = 0$ and its equation is $\mathbf{r} \cdot \mathbf{n} = 0$. If A, B, C are collinear, $\mathbf{n} = \mathbf{0}$ and the locus is only a line (or a single point if all three coincide), so the argument gives no information.

Checks

- With $\mathbf{a} = \mathbf{i}$, $\mathbf{b} = \mathbf{j}$, $\mathbf{c} = \mathbf{k}$, all three triple products equal 1, and $\mathbf{n} = \mathbf{i} + \mathbf{j} + \mathbf{k}$, $p = 1$: the plane $x + y + z = 1$, which indeed contains the three points.
- We tested 2000 random integer triples $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ with exact fractions: the expanded form of $\mathbf{n}$, the value of p, the equation $\mathbf{r} \cdot \mathbf{n} = p$ for random λ, μ, and the equality of the three triple products all held exactly. For coplanar triples every triple product was 0.

Common slips

- Sign errors when expanding $(\mathbf{b} - \mathbf{a}) \times (\mathbf{c} - \mathbf{a})$: the term $-\mathbf{b} \times \mathbf{a}$ becomes $+\mathbf{a} \times \mathbf{b}$, and $-\mathbf{a} \times \mathbf{c}$ becomes $+\mathbf{c} \times \mathbf{a}$.
- Forgetting to say why $\mathbf{n} \neq \mathbf{0}$, which is exactly where the non-coplanarity condition (via non-collinearity of A, B, C) is used.

Beyond STEP

The scalar triple product $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ is the 3×3 determinant with rows $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, and the cyclic symmetry proved here is the fact that a cyclic permutation of rows (two swaps) leaves a determinant unchanged. Its value is the signed volume of the parallelepiped on

the three vectors, and it vanishes precisely when they are linearly dependent, which is the coplanar case. In a first-year course this becomes the general theory of determinants as volume scale factors; Strang's *Introduction to Linear Algebra* develops it clearly.

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